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Week 3: Aperture and Resolution

Field Notes | Diffraction and Collecting Power
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Abstract

Aperture diameter is the first decisive optical parameter in an astronomical instrument because it simultaneously governs photon collection rate and diffraction-limited angular resolution. This journal develops both functions from first principles, treating aperture as a finite circular entrance pupil rather than a vague descriptor of telescope size. The collecting area relation shows that intercepted flux scales with the square of diameter, so a modest increase in aperture produces a disproportionately large gain in photon rate and shot-noise-limited signal-to-noise ratio. The distinction between aperture and magnification is established explicitly: magnification redistributes collected signal without increasing photon count or tightening the diffraction pattern. The wave-optical consequence of a finite pupil is then derived, establishing the Airy pattern, its energy distribution, and the Rayleigh criterion that connects wavelength and diameter to angular resolution. The exit pupil framework then links aperture to a physically grounded range of useful magnifications, bounding minimum and maximum for visual observation and identifying the empty magnification regime. A fully worked numerical example using the Askar 91F apochromatic (APO) refractor quantifies these outcomes at 550 nanometres (nm), with the Eta Carinae Nebula observed near the zenith from southern Victoria as the southern-hemisphere reference target. The journal then separates diffraction from atmospheric seeing and closes by placing aperture at Stage 4 of the Signal Chain Model.

Keywords: *aperture; diffraction; Rayleigh criterion; Airy disk; angular resolution; collecting area; exit pupil; magnification range; signal chain; atmospheric seeing; astrophotography; southern hemisphere; Austronomic*

Nomenclature

Symbol / Abbreviation	Definition
Airy FWHM	The full width at half maximum of the central lobe of the Airy diffraction pattern produced by a circular aperture: theta_FWHM approximately equals 1.028 lambda/D (Equation (3)). At 550 nm this evaluates to approximately 116.5 / D_mm in arcseconds, within 0.4 percent of the Dawes empirical limit (116 / D_mm). The Airy FWHM is the correct PSF width measure to compare against atmospheric seeing FWHM when assessing whether an instrument is diffraction-limited or seeing-limited. It is distinct from the Rayleigh criterion (1.22 lambda/D, Equation (2)), which is a two-point separability threshold and must not be used as a blur width.
APO	Apochromatic: a multi-element lens design in which three or more wavelengths are brought to a common focal point, suppressing residual chromatic aberration to below the detection threshold for imaging.
arcsec / arcsecond	One 3600th of one degree of arc (1 arcsec = pi / 648,000 rad = approximately 4.848 x 10^-6 rad). The standard unit for small angular separations in astronomical measurement.
arcminute	One 60th of one degree of arc (1 arcmin = 60 arcsec). Used for larger angular extents such as nebula diameters.
cm^2	Square centimetres. Unit of collecting area (1 cm^2 = 10^-4 m^2).
D	Clear aperture diameter of the optical instrument, in metres (m) unless stated otherwise. The unobstructed entrance pupil diameter.
Dec	Declination: the celestial coordinate measuring angular distance north (positive) or south (negative) of the celestial equator, expressed in degrees from -90 deg (south celestial pole) to +90 deg (north celestial pole). Analogous to geographic latitude.
empty magnification	The regime of telescope use beyond the maximum useful magnification M_max (Equation (4)), in which additional magnification enlarges the Airy diffraction disk without revealing any finer detail. At M_max the Airy disk has already been expanded to approximately four times the eye's 1-arcminute resolution limit; further magnification

Symbol / Abbreviation	Definition
	produces a larger but no more informative image. Empty magnification is a visual observing concept; its equivalent for camera-based detection depends on focal length, pixel scale, and integration time constraints treated in Weeks 4, 5, 8, and 9.
EP	Exit pupil: the image of the telescope objective formed by the eyepiece. The disc of collimated light that exits the eyepiece and must coincide with the observer's eye. Diameter: $EP = D / M$ (Equation (4)), in millimetres. Useful range: maximum EP approximately 7 mm (dark-adapted eye pupil limit; wider exit pupils waste aperture to the iris); minimum practical EP approximately 0.5 mm (below this, eye aberrations and floaters degrade the image). These two limits define M_min and M_max respectively.
FWHM	Full Width at Half Maximum: the width of an intensity profile measured at half its peak value. Two distinct FWHM quantities appear in this journal: (1) seeing FWHM, the width of the atmospheric seeing disk, used to characterise atmospheric turbulence at a site; (2) Airy FWHM (theta_FWHM approximately 1.028 lambda/D, Equation (3)), the width of the central lobe of the Airy diffraction pattern, used to characterise the telescope PSF. When determining whether an instrument is seeing-limited or diffraction-limited, seeing FWHM must be compared against Airy FWHM. The Rayleigh criterion (Equation (2)) is a separability threshold, not an FWHM measure, and must not be substituted for Airy FWHM in this comparison.
J2000	Standard astronomical coordinate epoch: 1 January 2000 at 12:00 Terrestrial Time (TT). All Right Ascension and Declination values in this journal are referenced to this epoch unless otherwise stated.
lambda	Wavelength of the observed electromagnetic radiation, in metres (m) in equations; in nanometres (nm) in prose and tables.
m	Metre: the SI base unit of length. Prefixes used: mm (millimetre, 10^-3 m), nm (nanometre, 10^-9 m).
mm	Millimetre: 10^-3 m. Used for aperture diameters and focal lengths throughout this journal.
M_max	Maximum useful magnification for visual observation: M_max = D_mm / 0.5 = 2 x D_mm. Derived from Equation (4) at the minimum practical exit pupil of 0.5 mm. Beyond M_max the image enters the empty magnification regime. At M_max, the Airy disk FWHM (Equation (3)) subtends approximately 3.9 arcminutes to the eye, roughly four times the eye's 1-arcminute resolution limit.
M_min	Minimum useful magnification for visual observation: M_min = D_mm / 7. Derived from Equation (4) at the maximum useful exit pupil of 7 mm, corresponding to the dark-adapted eye's maximum pupil diameter. Below M_min the exit beam is wider than the eye's entrance pupil and collected light is lost to the iris.
N	Regime parameter: the ratio D / r_0. When N is close to or below 1.0, the instrument approaches its diffraction limit. When N substantially exceeds 1.0, the delivered image quality is set by the atmosphere.
NGC	New General Catalogue: a catalogue of deep-sky objects (nebulae, galaxies, clusters) compiled by J. L. E. Dreyer in 1888 and used universally as a reference identifier in observational astronomy.
nm	Nanometre: 10^-9 m. The standard unit for optical wavelengths. Visible light spans approximately 380 nm (violet) to 700 nm (red).
OTA	Optical Tube Assembly: the telescope tube and its optics, considered as a unit excluding the mount, camera, focuser, and accessories.
PSF	Point Spread Function: the two-dimensional intensity pattern produced by an optical system in response to an ideal point source at infinity. For a circular, aberration-free aperture the PSF is the Airy pattern.
r0	Fried parameter: a measure of atmospheric coherence length, defined as the diameter of a circular aperture over which the wavefront phase variance is less than 1 rad^2 RMS. Larger r0 indicates calmer atmospheric conditions and finer delivered image quality. Units: metres.
RA	Right Ascension: the celestial coordinate measuring angular distance eastward along the celestial equator from the vernal equinox, expressed in hours (h), minutes (m), and seconds (s) from 0 h to 24 h. Analogous to geographic longitude.
Sep.	Angular separation between the two components of a double star system, measured in arcseconds. The value to compare against the instrument's theoretical resolution limit.
SNR	Signal-to-Noise Ratio: the ratio of the useful signal (photons detected from the target source) to the root-mean-square noise from all sources combined. Higher SNR produces cleaner, more detail-preserving images.
theta	Angular resolution. Two distinct quantities are defined in this journal: (1) theta_R = 1.22 lambda/D, the Rayleigh criterion (Equation (2)): the minimum angular separation at which two point sources are just resolvable as distinct, defined by the position of the first dark minimum of the Airy pattern. A separability threshold, not a blur width. Applicable to double-star resolution tests. (2) theta_FWHM approximately equals 1.028 lambda/D, the Airy disk FWHM (Equation (3)): the width of the PSF central lobe at half maximum. The correct quantity to compare against atmospheric seeing FWHM. Both are in radians in equations; when expressed in arcseconds the unit is stated explicitly.
WDS	Washington Double Star Catalog: the primary international reference catalogue for binary and multiple star systems, maintained by the U.S. Naval Observatory. The authoritative source for component separations and position angles used in this journal's experimental program.

1. Introduction

Week 2 established light as the information-bearing signal of astronomy and showed that, after geometric spreading, interstellar absorption, and atmospheric transmission, only a small fraction of source radiation survives to the telescope plane. Week 3 begins at the entrance pupil: aperture is the stage at which the system determines how much of that surviving wavefront is admitted into the instrument, and any signal not collected here is irrecoverable.

That boundary has two independent physical consequences. A larger aperture intercepts more of the incoming wavefront, so photon collection rate scales with aperture area, which is proportional to D^2. Any finite aperture also truncates the wavefront and causes diffraction, replacing the ideal point image with an interference pattern whose angular width scales as $1/D$ at fixed wavelength. Brightness and resolving power both originate at the aperture plane, and both improve with increasing diameter, though they scale differently.

A third point requires explicit treatment: aperture is not magnification. Magnification redistributes collected signal over a larger apparent scale; it does not increase photon count or tighten the diffraction pattern. The journal derives both the collecting area relation and the diffraction limit, applies them to a realistic southern-hemisphere instrument, and places aperture within the Signal Chain Model from Week 2.

2. Aperture as a Physical Parameter

Aperture diameter is the effective clear diameter through which electromagnetic radiation enters the optical system. For a refractor it is the clear diameter of the objective lens; for a Newtonian reflector it is the effective diameter of the primary mirror after subtracting the central obstruction of the secondary mirror support. The physically relevant quantity is not the nominal outer tube diameter but the unobstructed clear aperture, since only that portion of the boundary contributes to the photon budget.

Two physical outcomes are fixed at the aperture plane and cannot be recovered by any downstream component. The first is the photon count rate admitted into the instrument: this is set by the product of incident photon flux and collecting area, and no amplification downstream can supply photons that were never collected. The second is the maximum angular resolution the system can achieve in principle: wave optics applied to any finite aperture boundary produces diffraction that limits image sharpness regardless of optical quality, focuser precision, or processing applied after the fact.

Both outcomes depend on aperture diameter D, but they scale differently. Collecting area scales as D^2, so photon sensitivity grows steeply with relatively modest diameter increases. Diffraction-limited resolution scales as $1/D$ at fixed wavelength, so angular resolution improves linearly as diameter increases. These two scaling relations are derived in Sections 3 and 4 respectively.

- Photon collection rate scales as D^2 (derived in Section 3).
- Diffraction-limited angular resolution scales as $1/D$ at fixed wavelength (derived in Section 4).
- Shot-noise-limited signal-to-noise ratio scales linearly with D for equal integration time (derived in Section 3).
- Field of view at the detector is governed by focal length, not aperture (Week 4).

The geometric consequence of aperture diameter on intercepted collecting area is shown in Figure 1.

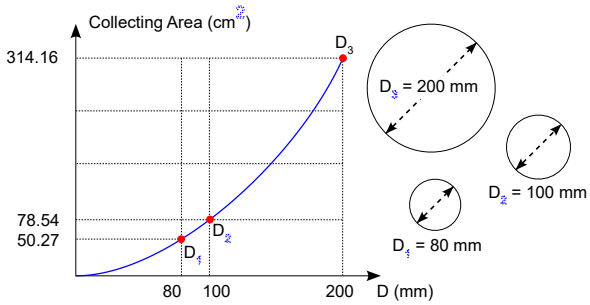

Figure 1. *Collecting area grows with the square of aperture diameter. Against a 100 mm reference, an 80 mm aperture intercepts 0.64 times the flux and a 200 mm aperture intercepts 4.00 times the flux. A linear increase in diameter produces a disproportionately larger increase in collected signal.*

3. Collecting Area and Signal Sensitivity

Magnification changes presentation scale, not admitted photon count or diffraction limit. A larger eyepiece magnification applied to a small aperture spreads the same photon budget over a larger apparent field without adding signal or sharpening detail. Aperture is the governing parameter at Stage 4 of the Signal Chain; all downstream stages operate on the photon budget that the entrance pupil admitted.

The photon count rate entering the telescope per unit time is the product of the incident photon flux at the aperture plane and the collecting area of the aperture. Incident flux is set by source luminosity and distance via the inverse-square law established in Week 2: a photon budget arrives at the instrument and the aperture determines what fraction of it is intercepted. For a circular unobstructed aperture, that fraction is proportional to D^2.

The shot-noise-limited signal-to-noise ratio follows from photon count statistics. Shot noise has a standard deviation equal to the square root of detected photon count N, so the signal-to-noise ratio (SNR) equals the square root of N. Since N is proportional to collecting area and collecting area is proportional to D^2, the SNR is proportional to the square root of D^2, which equals D. Signal-to-noise ratio scales linearly with aperture diameter for equal integration time, not with aperture area.

For a circular unobstructed aperture of diameter D, the collecting area is the area of the corresponding circle expressed in terms of diameter as given in Equation (1). Collecting area of a circular unobstructed aperture as a function of aperture diameter [2].

$$A = \pi \frac{D^2}{4} \tag{1}$$

A = collecting area of the aperture, in square metres (m^2)

D = clear aperture diameter, in metres (m)

The quadratic dependence in Equation (1) explains why aperture comparisons must use area ratios rather than diameter differences. Moving from 71 mm to 91 mm increases the linear dimension by 28 percent, but the area ratio is (91/71) squared, approximately 1.64: a 64 percent gain in intercepted flux. A target requiring an integration of 164 seconds at 71 mm to reach a given photon count would require approximately 100 seconds at 91 mm under otherwise identical conditions. To double the SNR for a fixed aperture requires quadrupling integration time, because SNR scales as the square root of total photons as shown in Equation (1) [2].

Table 1 quantifies collecting area and relative SNR for seven apertures normalised to a 100 mm reference, using Equation (1).

Table 1. *Collecting area and relative shot-noise-limited signal-to-noise ratio for circular apertures normalised to a 100 mm reference. SNR ratio assumes photon-noise-limited detection with equal integration time and sky conditions. Values are computed from Equation (1).*

Aperture *D* (mm)	Collecting Area (cm^2)	Area Ratio vs 100 mm	SNR Ratio vs 100 mm
71	39.6	0.50	0.71
80	50.3	0.64	0.80
91	65.0	0.83	0.91
100	78.5	1.00	1.00
130	132.7	1.69	1.30
150	176.7	2.25	1.50
200	314.2	4.00	2.00

For observers in southern Australia, the photon-rate advantage has direct consequences for extended targets such as the Eta Carinae Nebula (NGC 3372, where NGC is the New General Catalogue), which requires collecting sufficient signal from faint outer filamentary structure against the sky background. A modest aperture increase shifts the balance between required integration time and achievable depth before sky noise dominates.

4. Diffraction and the Point Spread Function

When a plane wave from a distant point source passes through a circular aperture, the wavefront is truncated at the boundary. According to the Huygens-Fresnel principle, every point on the truncated wavefront acts as a secondary source of spherical waves. The superposition of these waves at the focal plane produces an interference pattern rather than a geometric point image. For a circular aperture free of optical aberrations, this pattern is known as the Airy pattern: a bright central region, called the Airy disk, surrounded by alternating dark rings and progressively fainter bright rings. The Airy pattern is the impulse response of the optical system, referred to formally as the Point Spread Function.

The diffraction structure is not a consequence of manufacturing imperfection. It is an inescapable consequence of wave optics applied to any finite aperture boundary: no optical polish, alignment precision, or image processing removes it, because it arises from the physics of wavefront truncation, not from surface error. The only means of narrowing the Airy disk is to increase aperture diameter or to observe at a shorter wavelength. This constraint applies equally to the human eye, to camera lenses, and to the primary mirrors of professional observatories.

The energy within the Airy pattern is distributed across the central disk and outer rings in fixed proportions that follow from the Bessel function solution to the circular aperture diffraction integral. Table 2 gives the principal energy fractions for an ideal, aberration-free circular aperture [1].

Table 2. *Energy distribution within an ideal, aberration-free Airy pattern for a circular aperture. The central disk contains 83.8 percent of the diffracted energy. Proportions follow from the Bessel function solution to the diffraction integral and are independent of aperture diameter and wavelength.*

Region of Airy Pattern	Approximate Energy Fraction	Practical Implication
Central disk (inside first dark ring)	83.8%	Primary resolving core; governs the Rayleigh criterion
First bright ring	7.2%	Can interfere with faint close companions or tight double stars
Second bright ring	2.8%	Rarely detectable above sky background or read noise
Outer rings combined	6.2%	Contributes to a low-level halo in long-exposure imaging

4.1 The Rayleigh Criterion

To determine when two adjacent point sources remain distinguishable, a resolution threshold is required. The Rayleigh criterion specifies the angular separation at which the central maximum of one Airy pattern falls exactly on the first dark minimum of the adjacent pattern. At

this separation the combined intensity profile shows two distinct peaks with a dip of approximately 26.5 percent between them, sufficient for a detector to identify two separate sources. The criterion was formalised by Lord Rayleigh in 1879 and remains the standard engineering convention in optical and astronomical instrumentation. The minimum resolvable angular separation is given by Equation (2).

The Rayleigh criterion for the diffraction-limited angular resolution of a circular aperture, relating the minimum resolvable angular separation to wavelength and aperture diameter [1].

$$\theta \approx 1.22\frac{\lambda}{D} \qquad (2)$$

$\boldsymbol{\theta}$ = minimum resolvable angular separation, in radians (rad)

$\boldsymbol{\lambda}$ = wavelength of the observed radiation, in metres (m)

$\boldsymbol{D}$ = clear aperture diameter, in metres (m)

The coefficient 1.22 in Equation (2) is not a convenience approximation. It is the first zero of the Bessel function $J_1(x)$, equal to 3.8317, which appears in the full solution to the circular aperture diffraction integral as the factor 3.8317 divided by pi times (D divided by λ). A rectangular aperture of the same width produces a coefficient of 1.00 because its one-dimensional geometry yields a simpler integral. Equation (2) shows that at fixed wavelength, angular resolution improves inversely with diameter: doubling D halves the minimum resolvable angle. At 550 nm, a 71 mm aperture has a Rayleigh separation of approximately 1.95 arcseconds (where one arcsecond is 1/3600 of one degree, equivalent to approximately 4.848 x 10^-6 Rad), while a 91 mm aperture reaches approximately 1.52 arcseconds. Near the limit of resolution for close double stars or fine planetary structure, this difference determines whether the angular separation remains distinct or merges into a single broadened image as given in Equation (2) [1].

A critical distinction must be drawn before these values are applied to atmospheric comparisons. The Rayleigh criterion is a separability threshold: it defines the angular distance between two point sources at which a detector can first identify them as distinct, based on the position of the first dark minimum of the Airy pattern. It is not a measure of the width of the stellar image. Atmospheric seeing, by contrast, is characterised as a full width at half maximum (FWHM) of the seeing disk, a profile width measure. Placing the Rayleigh separation $\theta_R = 1.22\,\lambda/D$ directly alongside seeing FWHM conflates a separability criterion with a blur width: the two quantities originate from different definitions and are not equivalent. The correct PSF width measure to compare against seeing FWHM is the Airy disk FWHM: the width of the central lobe of the Airy pattern at half its peak intensity. For a circular aperture, this width is given by Equation (3).

The Airy disk full width at half maximum (FWHM) for a circular aperture: the PSF width measure to compare against atmospheric seeing FWHM when assessing the diffraction-limited versus seeing-limited regime [1].

$$\theta_{\mathrm{FWHM}} \approx 1.028\frac{\lambda}{D} \qquad (3)$$

$\boldsymbol{\theta_{\mathrm{FWHM}}}$ = full width at half maximum of the Airy disk central lobe, in radians (rad)

$\boldsymbol{\lambda}$ = wavelength of the observed radiation, in metres (m)

$\boldsymbol{D}$ = clear aperture diameter, in metres (m)

At 550 nm, Equation (3) evaluates to approximately 116.5 / D_{mm} in arcseconds. For a 71 mm aperture this gives an Airy FWHM of approximately 1.64 arcseconds; for a 91 mm aperture, approximately 1.28 arcseconds. Comparing these with the corresponding Rayleigh separations (1.95 and 1.52 arcseconds) reveals the Airy FWHM is systematically about 16 percent smaller, which is the direct consequence of the 1.028/1.22 ratio between the two coefficients. The empirical Dawes limit of 116 / D_{mm} closely tracks the Airy FWHM at 550 nm (116.5 / D_{mm}) to within 0.4 percent. This correspondence is not coincidental: Dawes derived his criterion from visual double-star observation, and the eye responds to the apparent width of the stellar image, not to the geometric position of a first minimum. Equation (2) therefore governs double-star separability in the experimental program of Section 8.2; Equation (3) governs the seeing versus diffraction comparison in Sections 6 and 8.1 [1, 2].

The physical relationship between finite aperture, diffraction, and the Rayleigh criterion is shown analytically in Figure 2 [1].

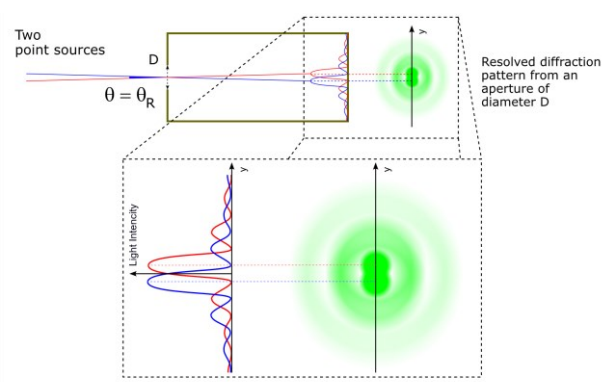

Figure 2. *The Rayleigh criterion illustrated analytically. Two point sources are at the resolution threshold when the central maximum of one Airy pattern falls at the first dark minimum of the other. The angular separation at this condition is theta_R = 1.22 lambda / D.*

Table 3 applies Equation (2) to five representative apertures at three wavelengths spanning the visible range, converting the equation into practical angular resolution values.

Table 3. *Theoretical diffraction-limited angular resolution via Equation (2) for five apertures at three representative wavelengths. Under typical seeing at southern Australian observing sites, atmospheric turbulence limits practical resolution to approximately 2 to 4 arcseconds, so the diffraction limit is the operative constraint only for small apertures or under exceptional conditions.*

Aperture D (mm)	Resolution at 400 nm (arcsec)	Resolution at 550 nm (arcsec)	Resolution at 700 nm (arcsec)
71	1.42	1.95	2.49
91	1.11	1.52	1.94
100	1.01	1.38	1.76
130	0.77	1.06	1.36
200	0.50	0.69	0.88

Table 3 makes explicit the wavelength dependence. At 91 mm aperture, the diffraction limit improves from 1.94 arcseconds at 700 nm to 1.11 arcseconds at 400 nm, a factor of 1.75 gain from observing in blue rather than red light. In practice, detector quantum efficiency, sky transparency, and target emission characteristics constrain which wavelength band is most productive for a given observation.

5. Worked Example: Askar 91F APO

The two quantitative relations derived in Sections 3 and 4 are now applied together to a commercially available southern-hemisphere instrument. The Askar 91F apochromatic refractor has a publicly documented clear aperture of 91 mm. The following three steps apply Equations (1) and (2) in sequence at a reference wavelength of 550 nm.

1. Collecting area via Equation (1), with D = 0.091 m: A = pi x (0.091)^2 / 4 = pi x 8.281 x 10^-3 / 4 = 6.50 x 10^-3 m^2 = **65.0 cm^2**.

2. Resolution limits via Equations (2) and (3), with lambda = 550 x 10^-9 m and D = 0.091 m. Rayleigh criterion (Equation (2), separability threshold): theta_R = 1.22 x (550 x 10^-9) / 0.091 = 7.37 x 10^-6 rad = 1.52 arcsec. This governs double-star resolution tests. Airy disk FWHM (Equation (3), PSF blur width): theta_FWHM = 1.028 x (550 x 10^-9) / 0.091 = 6.21 x 10^-6 rad = **1.28 arcsec**. This governs comparisons against atmospheric seeing FWHM. The two values are distinct by 16 percent and must not be substituted for one another.
3. Southern-hemisphere interpretation. The Eta Carinae Nebula (NGC 3372) transits within a few degrees of the zenith from southern Victoria (latitude approximately -37.9 degrees south). A zenith transit corresponds to an airmass of approximately 1.0, the minimum possible value, compared to airmass values of 1.5 to 2.0 that northern-hemisphere observatories incur when observing the same target at low elevation. Lower airmass reduces extinction losses at Stage 3 of the Signal Chain. The nebula spans approximately 120 arcminutes, far exceeding the 1.52 arcsecond diffraction limit; the primary aperture benefit for this target is therefore the 65.0 cm^2 collecting area, which reduces required integration time compared to smaller instruments.

For comparison, a 71 mm aperture yields a collecting area of 39.6 cm^2 and a diffraction limit of 1.95 arcseconds at 550 nm. The 91F APO provides an area ratio of 1.64 and a shot-noise-limited SNR advantage of 1.28 for equal integration time. For targets where angular resolution is the operative constraint, such as close double stars, the 91F's tighter diffraction limit of 1.52 arcseconds gives a further advantage over the smaller instrument.

6. Aperture in the Signal Chain and Atmospheric Limits

The two relations derived in this journal map onto the Signal Chain Model from Week 2. Stage 4 of that model is telescope collecting area, with aperture diameter *D* as its key controlling parameter. The analysis in Sections 3 and 4 gives this identification physical content: *D* governs the photon count rate via Equation (1) and the finest angular scale preserved in the collected wavefront via Equation (2). Both outcomes are fixed at Stage 4 and cannot be improved by any downstream element.

Diffraction is set by the telescope; seeing is imposed by the atmosphere. These are distinct physical phenomena that must not be conflated. Diffraction arises from wave optics applied to the finite aperture boundary and is a fixed property of the instrument that does not change with site conditions. Seeing arises from turbulent refractive-index variations along the line of sight through the atmosphere and varies with site, altitude, time, and meteorological conditions. When comparing these two limits, the correct pairing is seeing FWHM against Airy disk FWHM (Equation (3)), not against the Rayleigh criterion (Equation (2)). The Rayleigh criterion is a separability threshold; seeing FWHM is a blur width; only two quantities from the same class of definition may be meaningfully compared. At 550 nm, the 71 mm aperture has an Airy FWHM of approximately 1.64 arcseconds and the 91 mm aperture has an Airy FWHM of approximately 1.28 arcseconds. When seeing FWHM is 3 arcseconds and both instrument Airy FWHM values are substantially smaller, the atmosphere governs the delivered image width for both instruments equally: the resolution advantage of the larger aperture is not realised at all. Resolution advantage from aperture

requires seeing FWHM to be comparable to or finer than the instrument's Airy FWHM. Aperture still improves photon collection regardless of seeing, because collecting area is independent of atmospheric blur. The Fried parameter, which quantifies the atmospheric coherence length, is treated in Week 6.

- Increasing D improves photon collection as D^2 and angular resolution as $1/D$: both improve simultaneously.
- When seeing dominates, aperture still contributes its full collecting area gain, but the resolution advantage is suppressed until atmospheric conditions improve.
- For extended southern targets such as the Eta Carinae Nebula, collecting area is the primary aperture benefit under typical suburban seeing of 2 to 4 arcseconds.
- Image processing cannot recover angular detail that the aperture and atmosphere together failed to preserve at the focal plane.

Figure 3 places aperture within the full seven-stage signal chain and makes explicit the separation between the intrinsic diffraction limit and the external atmospheric limit. [3]

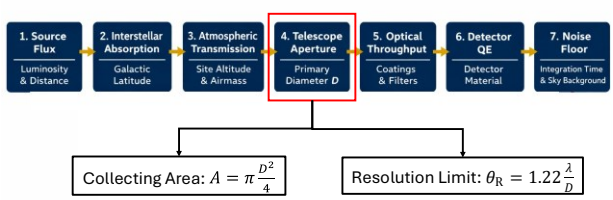

Figure 3. *Aperture occupies Stage 4 of the Signal Chain and simultaneously fixes photon collection rate and diffraction-limited resolution. Atmospheric seeing enters at Stage 3 as an independent, external limit distinct from the diffraction limit set by the aperture geometry.*

7. Summary and Bridge to Week 4

Aperture diameter governs two independent physical outcomes. Equation (1) shows that photon count rate scales as D^2: a 200 mm aperture collects four times the photons of a 100 mm aperture per unit time, yielding twice the shot-noise-limited signal-to-noise ratio for equal integration time. Equation (2) shows that diffraction-limited angular resolution scales as $1/D$: the 200 mm aperture resolves angular detail twice as fine at the same wavelength. Equation (4) closes the picture by showing that aperture also sets the useful magnification range for visual observation: $M_{\min} = D_{\mathrm{mm}} / 7$ and $M_{\max} = 2 \times D_{\mathrm{mm}}$ follow directly from the exit pupil geometry and the physiological limits of the human eye. The worked example for the Askar 91F APO confirmed a collecting area of 65.0 cm^2, a diffraction limit of 1.52 arcseconds at 550 nm, and a visual operating range of 13x to 182x. Observed from southern Victoria at near-zenith airmass, the Eta Carinae Nebula benefits primarily from collecting area, because its 120 arcminute extent exceeds all practical diffraction limits. Under typical suburban Australian seeing of 2 to 4 arcseconds, the atmosphere suppresses the resolution advantage of all apertures in the amateur range: the Fried parameter treatment in Week 6 establishes when aperture size produces a resolution gain as well as a sensitivity gain.

Week 4 moves from entrance-pupil diameter to image-formation geometry. Focal length determines how the wavefront collected at the aperture is projected onto the detector, setting image scale and field of view. Focal length does not alter the photon collection rate or the diffraction-limited resolution established here: those are fixed by D and λ at Stage 4. What focal length determines is how large a resolved source appears on the detector surface and how much sky fits within the sensor boundary. These are geometrically distinct from the aperture parameters developed in this journal and define the system-level trade-offs that are the subject of Week 4.

8. Instrument Trade Study: Aperture and Resolution

This section applies the resolution framework developed in Sections 4 and 4.1 to the full set of optical tube assemblies in the instrument collection. Two resolution criteria are evaluated for each instrument: the Rayleigh criterion from Equation (2) and the empirical Dawes limit. Both are compared against Melbourne's typical atmospheric seeing to identify which instruments can reach their diffraction limit from southern Victoria under realistic conditions. A visual magnification analysis using the exit pupil framework of Equation (4) then establishes the practical operating range for each instrument. An experimental program is then specified that uses double star pairs of known separation to test the resolution predictions observationally.

8.1 Analytical Methodology

Equation (2) gives the Rayleigh resolution limit in radians as $\theta = 1.22\,\lambda\,/\,D$. Converting to arcseconds at the reference wavelength of 550 nm and expressing aperture in millimeters yields the working form used in Table 4: $\boldsymbol{\theta_{\mathrm{R}} = 138\,/\,D_{\mathrm{mm}}}$. This derived form is exact for λ = 550 nm and follows directly from multiplying Equation (2) by the radian-to-arcsecond conversion factor of **206,265**, giving 1.22 x 550 x 10^-9 x 206,265 / D_{m} = 0.1383 / D_{m} = 138.3 / D_{mm}.

The Dawes limit is an empirical criterion derived from visual double-star observation by William Rutter Dawes. It represents the angular separation at which a skilled observer first detects elongation of the combined image, corresponding to an intensity dip of approximately 5 percent between the two maxima. It has no explicit wavelength dependence because it was determined **psychophysically from visual observation**. The working form is $\boldsymbol{\theta_{\mathrm{D}}(arcsec) = 116\,/\,D_{\mathrm{mm}}}$. The Rayleigh limit (138 / D_{mm}) is the physically grounded wave-optics criterion suitable for imaging; the Dawes limit (116 / D_{mm}) is the conventional standard for visual double-star work. Both are tabulated here because both appear in the literature and both are relevant depending on whether the instrument is used for imaging or visual observation.

A third criterion is required to complete the resolution framework: the Airy disk FWHM from Equation (3), $\boldsymbol{\theta_{\mathrm{FWHM}}} = 1.028\,\lambda\,/\,D_{\mathrm{mm}}$. At 550 nm this evaluates to $\boldsymbol{116.5\,/\,D_{\mathrm{mm}}}$ in arcseconds, within 0.4 percent of the Dawes limit. This near-identity confirms that the Dawes criterion, despite being derived empirically from visual observation, closely captures the Airy FWHM at visible wavelengths. The physical interpretation of each criterion is distinct and determines its correct application. **The Rayleigh criterion (138 / D_{mm}) is a separability threshold**: it predicts whether two adjacent point sources produce a minimum detectable intensity between their combined Airy patterns. It governs the double-star resolution tests in Section 8.2. The Airy FWHM (approximately 116.5 / D_{mm} at 550 nm, effectively equal to the Dawes values in Table 4) is a blur width: it characterises the angular extent of the PSF central lobe and is the correct quantity to compare against atmospheric seeing FWHM when determining whether an instrument is diffraction-limited or seeing-limited. **Comparing the Rayleigh criterion directly against seeing FWHM, as is common in popular astronomy literature, conflates a separability criterion with a blur width and overstates the aperture at which diffraction dominates by approximately 16 percent.** The regime column in Table 4 is therefore based on D/r_0 and the

Airy FWHM (Dawes) values, not the Rayleigh values.

A fourth analytical layer completes the methodology: the exit pupil (EP) framework, which governs the range of useful magnifications for **visual observation**. The exit pupil is the image of the telescope objective formed by the eyepiece: the disc of collimated light that emerges from the eyepiece and must be received by the observer's eye. By similar triangles in the afocal geometry of a Keplerian telescope, the exit pupil diameter is determined by Equation (4).

Exit pupil diameter as a function of aperture and angular magnification. EP and D are in the same units (typically millimetres); M is dimensionless [2, 4].

$$EP = \frac{D}{M} \tag{4}$$

$\boldsymbol{EP}$ = exit pupil diameter, in millimetres (mm)

$\boldsymbol{D}$ = clear aperture diameter, in millimetres (mm) for this working form

$\boldsymbol{M}$ = angular magnification, dimensionless

Equation (4) directly links aperture to a useful magnification range through two physiological limits of the human eye. The lower bound on magnification is set by the maximum useful exit pupil, which equals the dark-adapted eye's maximum pupil diameter of approximately 7 mm. When EP exceeds 7 mm, the exit beam is wider than the eye's entrance pupil and the surplus light falls on the iris rather than the retina; aperture is wasted and image brightness does not increase. Setting EP = 7 mm in Equation (4) gives the minimum useful magnification $M_{\text{min}} = D_{\text{mm}} / 7$. The upper bound is set by the minimum practical exit pupil, approximately 0.5 mm. Below this diameter, diffraction effects at the eye's own pupil edge become significant, eye floaters and optical aberrations become visible, and image quality degrades. Setting EP = 0.5 mm gives the maximum useful magnification $M_{\text{max}} = D_{\text{mm}} / 0.5 = 2 \text{ x } D_{\text{mm}}$. Beyond M_{max} the telescope enters empty magnification: the Airy disk is enlarged without any additional angular resolution being conveyed [2, 4].

The derivation of M_{max} from Equation (3) provides an independent physical cross-check. The Airy disk FWHM at 550 nm is 116.5 / D_{mm} arcseconds (Equation (3)). The human eye resolves approximately 60 arcseconds (1 arcminute) under dark-adapted conditions. For the Airy disk to be just perceptible as a resolved structure, M_{max} must expand theta_FWHM to at least 60 arcseconds: M_{max} = 60 / (116.5 / D_{mm}) = 0.515 x D_{mm}, or approximately D_{mm} / 2. This matches the exit pupil derivation to within the precision of the physiological constants. At M_{max} for a 71 mm aperture (M = 142), the Airy disk FWHM subtends approximately 233 arcseconds to the eye, about 3.9 arcminutes, which is nearly four times the eye's resolution limit. Magnification beyond this point enlarges a disk that the eye already resolves; it adds no new information. Table 4 includes M_{min} and M_{max} for each OTA as directly observable consequences of aperture [2].

The exit pupil framework applies specifically to visual observation, where the human eye is the detector. For camera-based imaging, the analogous bounds are governed by structurally parallel but physically distinct constraints: a minimum image scale set by the requirement to sample the PSF at sufficient angular resolution (determined by focal length and pixel pitch, treated in Weeks 4 and 8), and a practical upper bound on image scale set by the requirement that each resolution element receives sufficient photon signal within workable integration times (governed by focal ratio and sky background, treated in Weeks 5 and 9). The exit pupil

calculation here is the complete derivation for the visual regime; the imaging analog is deliberately deferred to the weeks whose content makes it derivable from first principles.

The atmospheric regime parameter $N = D/r_0$ determines whether a given aperture operates near its diffraction limit or is seeing-limited. When N is close to or below 1.0, the aperture is smaller than the atmospheric coherence length and the instrument can approach its diffraction limit. When N substantially exceeds 1.0, the atmosphere is the limiting factor and angular resolution is set by the seeing disk rather than by D. Melbourne's median atmospheric conditions correspond to a Fried parameter r_0 (the atmospheric coherence length, in metres) of approximately 70 mm (seeing full width at half maximum, FWHM, approximately 2.0 to 3.0 arcseconds). Table 4 presents the resolution and visual magnification parameters for all five optical tube assemblies (OTAs) in the instrument collection, using aperture values from the instrument datasheets.

The resolution limits in Table 4 are plotted as continuous curves in Figure 4 to show how both criteria vary across the full aperture range and where each instrument sits relative to Melbourne's median seeing floor [1, 2]. The key engineering result in Table 4 is not the absolute resolution values but the regime classification. The Askar 71F is the only OTA in the collection whose diffraction limit (1.94 arcseconds at 550 nm) lies within the range of Melbourne's typical seeing (2.0 to 3.0 arcseconds). Its D/r_0 ratio of 1.01 at median conditions places it exactly at the threshold between the diffraction-limited and seeing-limited regimes: on any night where r_0 reaches 71 mm or better, the 71F operates at its theoretical limit. This is a genuine optical advantage that a larger aperture does not share from this site under typical conditions.

Table 4. *Resolution and visual magnification parameters for all five OTAs in the instrument collection. Rayleigh limits (138 / D_{mm}, Equation (2) at 550 nm) are separability thresholds applicable to double-star tests. Dawes limits (116 / D_{mm}) correspond to the Airy disk FWHM (Equation (3), 116.5 / D_{mm} at 550 nm) to within 0.4 percent: these are the PSF blur widths for the seeing-regime comparison. Regime parameter $N = D/r_0$ evaluated at r_0 = 70 mm (Melbourne median). M_{min} = D_{mm} / 7 and M_{max} = 2 x D_{mm} are derived from Equation (4) at the 7 mm maximum and 0.5 mm minimum useful exit pupil respectively. All M values are rounded to the nearest integer.*

Instrument	Aperture D (mm)	Rayleigh θ_R (arcsec) at 550 nm	Dawes θ_D / Airy FWHM (arcsec) at 550 nm	D/r_0 at median seeing (r_0 = 70 mm)	Regime at median seeing	M_{min} ($D/7$)	M_{max} ($2D$)
Askar 71F	71	1.94	1.63	1.01	Threshold: near diffraction limit on median nights	10x	142x
Askar 91F APO	91	1.52	1.27	1.30	Mildly seeing-limited on most nights	13x	182x
Askar 103APO	103	1.34	1.13	1.47	Seeing-limited on all but the best nights	15x	206x
EdgeHD 800	203	0.68	0.57	2.90	Strongly seeing-limited under typical conditions	29x	406x
EdgeHD 9.25	235	0.59	0.49	3.36	Strongly seeing-limited under typical conditions	34x	470x

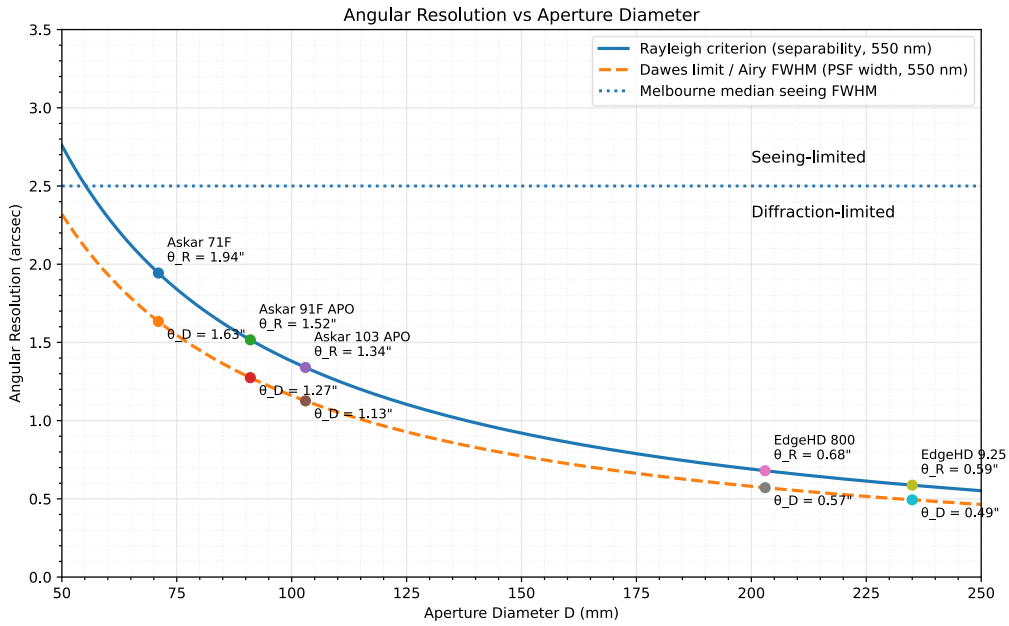

Figure 4. *Rayleigh criterion (solid) and Dawes limit / Airy FWHM (dashed) plotted continuously across the aperture range of the instrument collection. The horizontal dashed grey line at 2.5 arcseconds represents Melbourne's median seeing FWHM. The regime boundary is assessed by comparing seeing FWHM against the Airy FWHM (dashed, Equation (3)), not the Rayleigh criterion (solid, Equation (2)). Only the Askar 71F (D/r0 = 1.01) regularly approaches its diffraction limit from southern Victoria under median conditions.*

The visual magnification range of 10x to 142x covers all practical eyepiece combinations and presents the smallest $M_{\max}$ of any instrument in the collection, an appropriate match for a compact, fast refractor.

The EdgeHD 800 and EdgeHD 9.25 present the counter-intuitive result central to this trade study. Their theoretical diffraction limits of 0.68 and 0.59 arcseconds are the finest in the collection, yet their D/r_0 values of 2.90 and 3.36 mean that the atmosphere blurs their images to approximately 2.5 arcseconds on a median night, the same seeing disk that limits the smaller instruments. The resolution advantage of the EdgeHD instruments is inaccessible from southern Victoria under typical conditions. Their value in the collection is photon collection rate and fine image scale for compact targets on the best nights of the year, not superior angular resolution on typical nights. This conclusion follows directly from the same Equation (2) and aperture values that predict their impressive diffraction limits: the physics is self-consistent. Their $M_{\max}$ values of 406x and 470x reflect the larger aperture and would only be reached under the exceptional sub-arcsecond seeing needed to access their diffraction limits. Any content that presents the EdgeHD instruments as routinely resolving more detail than the 91F under average Melbourne seeing (2~3 arcsec) would be physically incorrect.

8.2 Experimental Program

Objective: to confirm that each instrument achieves, on a real night, the angular resolution that its aperture and the atmospheric conditions on that night jointly predict. Specifically, the experiment tests whether the Askar 71F can cleanly resolve a double star pair whose separation matches its predicted diffraction limit

when seeing is at or below 2.0 arcseconds, and whether the EdgeHD instruments fail to show the resolution improvement their diffraction limits would imply under the same typical seeing conditions. Confirmation criterion: measured separation of a resolved pair within 5 percent of the catalogued separation from the Washington Double Star Catalog [6].

8.2.1 Target Selection

Five double star targets are specified in order of increasing difficulty. The sequence is intentional: the experiment begins where resolution is certain for all instruments and progresses to separations where success depends on atmospheric conditions and aperture. Only pairs with magnitude difference below 1.5 are used; unequal pairs are excluded because the bright primary creates a halo that can prevent detection of the faint secondary even when the angular gap is above the diffraction limit. Table 5 gives the full target specifications. Separations for Alpha Centauri AB and Gamma Centauri must be confirmed from the Washington Double Star Catalog before each session, as both are orbital binaries whose separations change from year to year.

Column headers in Table 5 use standard astronomical notation. RA is Right Ascension, the celestial equivalent of longitude, measured eastward in hours (h) and minutes (m) from 0 h to 24 h along the celestial equator. Dec is Declination, the celestial equivalent of latitude, measured in degrees north (positive) or south (negative) of the celestial equator. Both coordinates carry the epoch designation J2000, meaning they are referenced to the standard astronomical reference frame of 1 January 2000 at 12:00 Terrestrial Time; coordinates shift slowly over decades due to precession and must be epoch-stamped for precision work. Sep. is the angular separation between the two components of the double star, measured in arcseconds, and is the quantity compared directly against the instrument's Rayleigh resolution limit.

Table 5. *Double star resolution test sequence for the Week 3 experimental program. Targets are ordered by increasing difficulty. Transit altitudes computed for latitude -37.9 degrees south (southern Victoria) using h = 90 deg minus $|\phi - \delta|$. Alpha Centauri AB and Gamma Centauri separations are orbital; confirm current values from the Washington Double Star Catalog [6] before each session.*

Target	RA (J2000) h m	Dec (J2000) deg	Sep. (arcsec)	Magnitudes	Transit alt. (lat -37.9 deg S)	Best months	Role in session
Theta Eridani (Acamar)	02h 58m	-40.3 deg	8.3	3.2 / 4.3	~88 deg	Oct to Jan	Session calibration: confirms optics and atmosphere are stable before proceeding
Acrux (alpha Crucis)	12h 27m	-63.1 deg	4.0	1.3 / 1.7	~65 deg	Mar to Jun	Primary test: above all instrument Rayleigh limits; clean unbiased test for all apertures
Alpha Centauri AB	14h 40m	-60.8 deg	~8 to 10 (orbital)	0.0 / 1.3	~67 deg	Mar to Jun	Standard reference: brightest southern double; confirm current separation from WDS before session
Zeta Aquarii	22h 29m	-0.0 deg	2.1	4.4 / 4.6	~52 deg	Aug to Oct	Demanding test: between Rayleigh limits of 91F (1.52) and 71F (1.94); tests 103APO and larger only
Gamma Centauri	12h 42m	-48.9 deg	~1.3 (orbital)	2.9 / 2.9	~79 deg	Mar to Jun	Stress test: below Rayleigh limit of all plan instruments; requires seeing below 1.3 arcsec

8.2.2 Observing Site Guidance

The resolution test requires steady atmospheric seeing. Sites with low thermal gradients (away from roads, buildings, and recently mown grass) and unobstructed southern horizon access perform best. The following four locations within 60 to 90 minutes of metropolitan Melbourne, Victoria are suitable for mounted sessions with any instrument in the collection:

- Heathcote / Lake Eppalock area (approximately 90 min northeast of Melbourne CBD via the McIvor Highway). Bortle class approximately 4. Open flat terrain adjacent to the Lake Eppalock reservoir provides unobstructed views to all horizons including due south. No formal access restriction on public land adjacent to the reserve. Thermal stability is good after sunset when the reservoir moderates local temperature. Southern horizon: unobstructed to the ground horizon.
- Lancefield / Romsey agricultural area (approximately 60 min north of Melbourne CBD via the Calder Freeway). Bortle class approximately 5. Flat agricultural land on public rural roadsides provides good all-horizon access. Light pollution from Melbourne is detectable to the south, limiting sky background but not significantly affecting stellar resolution measurements. Suitable for mounted sessions on stable ground.
- Bunyip State Forest fringe (approximately 65 min east-southeast of Melbourne CBD via the Princes Highway). Bortle class approximately 4 to 5 in open clearings near the forest perimeter. Access roads are unsealed; a standard vehicle can reach suitable clearings in dry conditions. Tree line limits the horizon below approximately 10 degrees altitude on the eastern and western sides; the southern horizon is clear from open areas. Good for sessions where target transit altitude is above 30 degrees.
- Meredith / Bannockburn agricultural area (approximately 75 min west of Melbourne CBD via the Western Ring Road and Western Freeway). Bortle class approximately 4 to 5. Very flat terrain provides excellent southern and eastern horizon access. Public rural roadsides are accessible year-round. Wind exposure on flat land can affect guiding; a natural windbreak such as a tree line or building should be used if available.

8.2.3 Procedure

The following steps apply to a complete resolution test session using the Askar 71F + ASI585MC on any equatorial mount, controlled via ASIAIR Plus. Where a different OTA from the collection is used, substitute that instrument and camera at every step where the 71F is named, and apply the corresponding plate scale for the new combination.

4. Thermal equilibration. Uncover the Askar 71F at least 30 minutes before observing begins. Do not begin the resolution test until the tube has reached ambient temperature. A warm objective produces a characteristic blurring in defocused star images that mimics poor seeing and invalidates the resolution test.
5. Defocused star test. Before any resolution test, point the 71F at an isolated bright star (Canopus, RA 06h 24m, Dec -52.7 deg, or Achernar, RA 01h 38m, Dec -57.2 deg). In the ASIAIR Plus live view, defocus the star by moving the focuser inward approximately 20 steps past focus. Both the intra-focal and extra-focal ring patterns should appear concentric and symmetric. Any asymmetry indicates collimation error or a thermal

gradient still present in the optics. Do not proceed to the resolution test until the star test rings are clean.

6. Calibration target. Point to Theta Eridani (RA 02h 58m, Dec -40.3 deg). Confirm a clean resolution with a visible dark gap between the two components at 8.3 arcsecond separation. If the pair cannot be cleanly split, the session is not suitable for resolution testing: either seeing is too poor or the optics have not equilibrated. Record the observation result and the time.
7. Seeing measurement. Before progressing to the primary test targets, capture 20 frames of 30 seconds on a rich star field (outer halo of Omega Centauri, RA 13h 27m, Dec -47.5 deg, if available). In the ASIAIR Plus, use the Auto Focus result or a manual SubframeSelector run in PixInsight after the session to measure the median FWHM in arcseconds. Record this as the session seeing. If FWHM exceeds 3.5 arcseconds, skip Zeta Aquarii and Gamma Centauri.
8. Primary resolution test. Point to Acrux (RA 12h 27m, Dec -63.1 deg). In the ASIAIR Plus, capture 100 to 200 frames at 50 to 200 millisecond exposure with the ASI585MC at high gain. This is a lucky imaging sequence: short enough to freeze atmospheric motion. Transfer the frames to AutoStakkert or AstroSurface on the connected laptop. Stack the best 10 percent of frames. In the stacked result, measure the separation in pixels between the two components using PixInsight's Measurement tool. Convert to arcseconds using the plate scale for the active instrument and camera combination (2.31 arcsec/pixel for the Askar 71F + ASI585MC at native focal length). Compare to the catalogued 4.0 arcsecond separation from the Washington Double Star Catalog [6]. Agreement within 5 percent confirms both the plate scale and the resolution result.
9. Graduated difficulty sequence. Repeat step 5 for Alpha Centauri AB (current separation from WDS [6]), then Zeta Aquarii (2.1 arcsec), and finally Gamma Centauri (current separation from WDS [6]) only if session seeing FWHM is below 1.5 arcseconds. For each target, record: whether the pair is cleanly resolved with a dark gap (confident resolution), resolved with a notch only (threshold), elongated without a notch (approaching limit), or unresolved. Record the time and the contemporaneous FWHM from SubframeSelector.
10. Data recording. For each target, enter in the session log: the instrument and camera combination used, the measured FWHM at time of observation, the plate-solved image scale (from ASIAIR Plus plate solve or Astrometry.net), the measured separation in arcseconds, the catalogued separation, the percentage discrepancy, and the resolution classification from step 6.

8.2.4 Expected Results and Discrepancy Criteria

Under session seeing of 2.0 arcseconds FWHM: Theta Eridani (8.3 arcsec) should resolve cleanly with all instruments. Acrux (4.0 arcsec) should resolve cleanly with all instruments. Zeta Aquarii (2.1 arcsec) should not be cleanly resolved by the 71F (Rayleigh limit 1.94 arcsec) but should be resolved by the 91F APO (1.52 arcsec) and all larger apertures on a good night. Gamma Centauri (approximately 1.3 arcsec) should not be resolved by any instrument in this collection under 2.0 arcsecond seeing; resolution of this target requires seeing below 1.3 arcseconds.

Measured separation agreement within 5 percent of the catalogued value constitutes

confirmation. Discrepancies greater than 5 percent most commonly arise from: incorrect plate scale due to a reducer not at its designed back-focus distance (not applicable to the native focal length configuration of the Askar 71F + ASI585MC, which carries no reducer); a stacking selection threshold that is too generous, including frames with motion blur; or a seeing measurement taken at a different time from the resolution test. If Acrux cannot be resolved cleanly with the 71F on a night where FWHM is measured at 2.0 arcseconds or better, investigate the optical train for a thermal gradient, collimation error, or focus drift.

8.2.5 Multi-Instrument Comparison Methodology

Comparing resolution performance across instruments requires care, because night-to-night seeing variation is a significant confound. A measurement made with the 71F on one night and the 91F APO on a different night cannot reliably isolate the aperture effect: a 0.3 to 0.5 arcsecond shift in seeing between the two sessions is sufficient to push an instrument across the diffraction-limited threshold, reversing the apparent outcome.

The minimum valid basis for a multi-instrument comparison is a sequential same-night session: both instruments are set up at the same site and the resolution test is repeated with each on the same target within a short window. The session seeing measurement (step 4 in the procedure) is then approximately common to both measurements, making the comparison more meaningful. The Acrux test is the first target to use for this purpose because its 4.0 arcsecond separation lies above the Rayleigh limit of all instruments in the collection and provides a reliable resolved reference.

The rigorous standard for a valid comparison eliminates night-to-night variation entirely by mounting both instruments simultaneously and imaging the same target concurrently. This mode is applicable when the available mount supports a dual-instrument configuration. With simultaneous data, any difference in resolved separation between the two instruments is attributable to the instruments themselves rather than to changing atmospheric conditions. This is particularly important for comparing instruments with D/r0 values near 1.0, such as the 71F (D/r_0 = 1.01), where a small improvement in seeing on a sequential night could produce a false impression of superior resolution from the second instrument.

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